The Multivariate Operator Alzer Inequality via a Löwner-Monotone Complement Path

Candidate full solution to the open problem in arXiv:1806.10806

17 August 2026

Status. Candidate full solution, likely valid pending expert review. The proof covers arbitrary (not necessarily commuting) positive operators and also gives the equality case.

1 Source problem

Morassaei and Mirzapour ask on pages 5–6 of [1] whether, for $0 < A_j \le \frac{1}{2} I$, the recursively ordered geometric mean satisfies

$$\mathbf{A}_n' - \mathbf{G}_n' \ \le \ \mathbf{A}_n - \mathbf{G}_n. \tag{1}$$

Here $A_j' = I - A_j$, $\mathbf{A}_n = n^{-1}\sum_j A_j$, and primes on the means mean that they are evaluated at $(A_1', \dots, A_n')$. The source's geometric mean is

$$X \#_\alpha Y = X^{1/2}(X^{-1/2} Y X^{-1/2})^\alpha X^{1/2},$$
$$\mathbf{G}_1(A_1) = A_1, \qquad \mathbf{G}_m(A_1, \dots, A_m) = A_1 \#_{(m-1)/m} \mathbf{G}_{m-1}(A_2, \dots, A_m).$$

The two images below reproduce the complete statement.

symmetric, but $\mathbf{G}_m$ is not symmetric for $m \ge 3$, for more details see [11, Example 2.3].

The geometric operator mean $\mathbf{G}_m(A_1, A_2, \cdots, A_m)$ has nice properties that for seeing more details c.f. [11].

Open Problem. Let $A_1, \cdots, A_n$ be n selfadjoint operators on an Hilbert space $\mathcal{H}$ such that $0 < A_j \le \frac{1}{2} I$, where I is identity operator on $\mathcal{H}$ [6, 8]. Also, let $\mathbf{A}_n := \mathbf{A}_n(A_1, \cdots, A_n)$ and $\mathbf{G}_n := \mathbf{G}_n(A_1, \cdots, A_n)$ be arithmetic and geometric means of $A_1, \cdots, A_n$ [11], and $\mathbf{A}_n' := \mathbf{A}_n(A_1', \cdots, A_n')$ and $\mathbf{G}_n' := \mathbf{G}_n(A_1', \cdots, A_n')$ be arithmetic and geometric means of $A'_1, \cdots, A'_n$ where $A'_j := I - A_j \quad (j = 1, \cdots, n)$, respectively. Then it seems that

$$\mathbf{A}'_n - \mathbf{G}'_n \leq \mathbf{A}_n - \mathbf{G}_n.$$

2 Proof intuition

Join each A_j to its complement by

$$A_j(t) = A_j + t(I - 2A_j), \qquad 0 \leq t \leq 1.$$

All directions $I - 2A_j$ are positive, and every path passes through the same point $I/2$ at $t = 1/2$. Joint monotonicity and direct-sum covariance of the recursive geometric mean imply something stronger than ordinary monotonicity: every scalar quadratic form of $F(t) = \mathbf{G}_n(A_1(t), \ldots, A_n(t))$ is an operator-monotone scalar function of t. Löwner's integral representation then says that its secant over $[0, 1]$ dominates its derivative at $1/2$. At the common diagonal point, the derivative of $\mathbf{G}_n$ is exactly the arithmetic average of the input directions. This turns the secant inequality into (1) after one rearrangement. A second-order expansion at the diagonal also yields the equality characterization.

3 Two one-variable lemmas

Lemma 1 (midpoint secant inequality)**.** *Let $f : [0, \infty) \to [0, \infty)$ be operator monotone and continuous at zero. Then*

$$f(1) - f(0) \ \geq \ f'(1/2). \tag{2}$$

Equality holds only if f is affine on $(0, \infty)$.

Proof. The Löwner representation for a nonnegative operator-monotone function on the positive half-line can be written

$$f(t) = a + bt + \int_{(0,\infty)} \frac{t}{t + \lambda} \, d\mu(\lambda), \qquad a, b \geq 0,$$

after absorbing harmless positive normalizing factors into the positive measure; continuity at zero absorbs a possible zero atom into a. For the nonlinear kernel $k_\lambda(t) = t/(t + \lambda)$,

$$k_\lambda(1) - k_\lambda(0) - k'_\lambda(1/2) = \frac{1}{1 + \lambda} - \frac{\lambda}{(\lambda + 1/2)^2} = \frac{1}{4(1 + \lambda)(\lambda + 1/2)^2} > 0.$$

The affine part gives equality. Integration proves (2); equality forces the representing measure to vanish. $\square$

Lemma 2 (positive multivariate paths become operator monotone)**.** *Let M be a positive operator map of n positive variables which is jointly monotone, respects orthogonal direct sums, and is unitarily covariant. If $P_j > 0$ and $Q_j \geq 0$, set*

$$F(t) = M(P_1 + tQ_1, \dots, P_n + tQ_n), \qquad t \geq 0.$$

For every unit vector $u \in \mathcal{H}$, the scalar function $f_u(t) = \langle F(t)u, u\rangle$ is nonnegative and operator monotone on $[0, \infty)$.

Proof. Fix $r \geq 1$ and positive matrices $S \leq T$ in M_r. On $\mathcal{H} \otimes \mathbb{C}^r$, put

$$\widehat{F}(S) = M(P_1 \otimes I + Q_1 \otimes S, \dots, P_n \otimes I + Q_n \otimes S).$$

Joint monotonicity gives $\widehat{F}(S) \leq \widehat{F}(T)$. If $S = U \operatorname{diag}(s_1, \dots, s_r)U^*$, direct-sum covariance and unitary covariance give

$$\widehat{F}(S) = (I \otimes U) \operatorname{diag}(F(s_1), \dots, F(s_r))(I \otimes U^*).$$

Compressing by the isometry $V : \mathbb{C}^r \to \mathcal{H} \otimes \mathbb{C}^r$, $Ve_k = u \otimes e_k$, yields

$$f_u(S) = V^* \widehat{F}(S) V \leq V^* \widehat{F}(T) V = f_u(T),$$

where the outer occurrences of f_u are ordinary scalar functional calculus. This holds for every r, hence f_u is operator monotone. □

The recursive mean $\mathbf{G}_n$ satisfies the three hypotheses of Lemma 2: weighted Kubo–Ando geometric means are monotone in both variables, direct-sum preserving, and unitarily covariant, and these properties pass through the recursion.

4 Full solution

Lemma 3 (derivative at a scalar diagonal)**.** *For selfadjoint $H_1, \dots, H_n$ and $c > 0$,*

$$\left.\frac{d}{ds}\right|_{s=0} \mathbf{G}_n(cI + sH_1, \dots, cI + sH_n) = \frac{1}{n}\sum_{j=1}^{n} H_j.$$

Proof. At (cI, cI) the first derivative of $X \#_\alpha Y$ in directions (H, K) is $(1-\alpha)H + \alpha K$. This follows immediately by expanding the defining formula to first order. Apply it to the recursion with $\alpha = (n-1)/n$ and use induction. □

Theorem 4 (multivariate operator Alzer inequality)**.** *Let $n \geq 2$ and let $A_1, \dots, A_n \in \mathbb{B}(\mathcal{H})$ satisfy $0 < A_j \leq \frac{1}{2}I$. For the recursively ordered geometric mean in [1],*

$$\boxed{\mathbf{A}'_n - \mathbf{G}'_n \ \leq\ \mathbf{A}_n - \mathbf{G}_n.}$$

No commutativity hypothesis is needed. Equality holds if and only if $A_1 = \cdots = A_n$.

Proof. Put $Q_j = I - 2A_j \geq 0$ and

$$F(t) = \mathbf{G}_n(A_1 + tQ_1, \dots, A_n + tQ_n), \qquad t \geq 0.$$

Thus $F(0) = \mathbf{G}_n$, $F(1) = \mathbf{G}'_n$, and every input at $t = 1/2$ equals $I/2$. Lemmas 1 and 2, applied to every unit vector u, imply

$$\langle (F(1) - F(0) - F'(1/2))u, u\rangle \geq 0.$$

Lemma 3 gives

$$F'(1/2) = \frac{1}{n}\sum_{j=1}^{n} Q_j = I - 2\mathbf{A}_n.$$

Consequently

$$\mathbf{G}'_n - \mathbf{G}_n \geq I - 2\mathbf{A}_n,$$

which is equivalent to $I - \mathbf{A}_n - \mathbf{G}'_n \leq \mathbf{A}_n - \mathbf{G}_n$, namely the asserted inequality.

It remains to identify equality. If equality holds, then equality holds in Lemma 1 for every scalarization, so $F''(1/2) = 0$. We show that this forces all Q_j to agree. The second-order expansion of a weighted geometric mean at a scalar diagonal is

$$(X\#_\alpha Y)''\big|_0 = (1-\alpha)X''(0) + \alpha Y''(0) - \frac{\alpha(1-\alpha)}{c}\big(X'(0) - Y'(0)\big)^2, \tag{3}$$

when $X(0) = Y(0) = cI$. (Expand the defining formula to second order.) Apply (3) recursively at $c = 1/2$. If $\overline{Q}_{2:n} = (n-1)^{-1}\sum_{j=2}^{n} Q_j$ and $\alpha = (n-1)/n$, then

$$F''_n(1/2) = \alpha F''_{n-1}(1/2) - \frac{\alpha(1-\alpha)}{c}(Q_1 - \overline{Q}_{2:n})^2.$$

Inductively $F''_{n-1}(1/2) \leq 0$. A sum of two negative operators can vanish only if both vanish. Hence $Q_1 = \overline{Q}_{2:n}$ and, recursively, $Q_2 = \cdots = Q_n$; therefore all Q_j, and all A_j, are equal. The converse is immediate because both arithmetic–geometric gaps then vanish. □

5 Verification and scope

The proof is dimension-free and does not use the arithmetic–geometric upper bound. Its essential review points are: (i) the amplification and compression in Lemma 2; (ii) the standard positive-half-line Löwner representation; and (iii) the diagonal derivative in Lemma 3. The equality statement additionally uses only the second-order identity (3).

Independent numerical tests in the accompanying code evaluated the exact recursive mean by spectral functional calculus on large random collections of real and complex positive matrices, including $n = 3, 4$ and dimensions $2, 3$, and a global/local optimization over real 2×2 triples. No negative eigenvalue beyond floating-point noise was found. These tests are sanity checks, not part of the proof.

The same argument proves a broader principle: any jointly monotone, direct-sum preserving, unitarily covariant multivariate mean has the complement inequality associated with its tangent weights at a scalar diagonal. The present theorem specializes those weights to $1/n$.

6 Novelty check

A bounded search on 17 August 2026 covered the run's cheap indexes and local arXiv source corpus, exact-title/author searches for arXiv:1806.10806, and web searches for "multivariate operator Alzer inequality", "operator Ky Fan type inequality", and recursive/multivariate geometric-mean variants. The later paper [2] cites the source and proves the arbitrary noncommuting *binary* Kubo–Ando-mean inequality, but it does not treat the recursive n-variable problem. Hansen [3] develops the regularity and monotonicity framework for inductive multivariate geometric means, but no complement inequality of the form above was located. Thus novelty confidence is moderate pending a specialist search.

7 Human review recommendation

Prioritize a check that the direct-sum calculation in Lemma 2 really identifies the compression with scalar functional calculus for every matrix level. Then check the normalization of the Löwner kernel in Lemma 1 and the convention $\mathbf{G}_m = A_1 \#_{(m-1)/m} \mathbf{G}_{m-1}$. The main inequality does not depend on the equality-case Hessian calculation, so any issue confined to (3) would affect only the characterization of equality.

References

[1] A. Morassaei and F. Mirzapour, *Alzer Inequality for Hilbert Spaces Operators*, arXiv:1806.10806 (2018).

[2] S. Habibzadeh, J. Rooin, and M. S. Moslehian, *Operator Ky Fan type inequalities*, arXiv:1811.00475 (2018), especially Theorem 3.1.

[3] F. Hansen, *Regular operator mappings and multivariate geometric means*, arXiv:1403.3781 (2014).

[4] R. Bhatia, *Positive Definite Matrices*, Princeton University Press, 2007.